

# Two-way fixed effects estimators with heterogeneous treatment effects\*

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## Abstract

Linear regressions with period and group fixed effects are widely used to estimate treatment effects. We show that they estimate weighted sums of the average treatment effects (ATE) in each group and period, with weights that may be negative. Due to the negative weights, the linear regression coefficient may for instance be negative while all the ATEs are positive. We propose another estimator that solves this issue. In the two applications we revisit, it is significantly different from the linear regression estimator.

**Keywords:** linear regressions, fixed effects, heterogeneous treatment effects, difference-in-differences.

**JEL Codes:** C21, C23

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# 1 Introduction

A popular method to estimate the effect of a treatment on an outcome is to compare over time groups experiencing different evolutions of their exposure to treatment. In practice, this idea is implemented by estimating regressions that control for group and time fixed effects. Hereafter, we refer to those as two-way fixed effects (FE) regressions. We conducted a survey, and found that 20% of all empirical articles published by the American Economic Review (AER) between 2010 and 2012 have used a two-way FE regression to estimate the effect of a treatment on an outcome. When the treatment effect is constant across groups and over time, such regressions estimate that effect under the standard “common trends” assumption. However, it is often implausible that the treatment effect is constant. For instance, the minimum wage’s effect on employment may vary across US counties, and may change over time. This paper examines the properties of two-way FE regressions when the constant effect assumption is violated.

We start by assuming that all observations in the same  $(g, t)$  cell have the same treatment and that the treatment is binary, as is for instance the case when the treatment is a county-level law. We consider the regression of  $Y_{i,g,t}$ , the outcome of unit  $i$  in group  $g$  at period  $t$  on group fixed effects, period fixed effects, and  $D_{g,t}$ , the treatment in group  $g$  at period  $t$ . Let  $\widehat{\beta}_{fe}$  denote the coefficient of  $D_{g,t}$ , and let  $\beta_{fe}$  denote its expectation. Under the common trends assumption, we show that  $\beta_{fe}$  is equal to a weighted sum of the treatment effect in each treated  $(g, t)$  cell:

$$\beta_{fe} = E \left( \sum_{(g,t): D_{g,t}=1} W_{g,t} \Delta_{g,t} \right). \quad (1)$$

$\Delta_{g,t}$  is the average treatment effect (ATE) in group  $g$  and period  $t$  and the weights  $W_{g,t}$ s sum to one but may be negative. Negative weights arise because  $\widehat{\beta}_{fe}$  is a weighted sum of several difference-in-differences (DID), which compare the evolution of the outcome between consecutive time periods across pairs of groups. However, the “control group” in some of those comparisons may be treated at both periods. Then, its treatment effect at the second period gets differenced out by the DID, hence the negative weights.

The negative weights are an issue when the ATEs are heterogeneous across groups or periods. Then, one could have that  $\beta_{fe}$  is negative while all the ATEs are positive. For instance,  $1.5 \times 1 - 0.5 \times 4$ , a weighted sum of 1 and 4, is strictly negative. Using the data set of Gentzkow et al. (2011), we find that 40% of the weights attached to  $\beta_{fe}$  are negative, so  $\beta_{fe}$  is not robust to heterogeneous effects.<sup>1</sup>

Researchers may want to know how serious that issue is in the application they consider. We show that conditional on all treatments, the absolute value of the expectation of  $\widehat{\beta}_{fe}$  divided by

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<sup>1</sup> Gentzkow et al. (2011) do not estimate  $\beta_{fe}$ , but  $\beta_{fd}$ , the treatment coefficient in the first-difference regression defined below. 46% of the weights attached to  $\beta_{fd}$  are strictly negative.

Our paper is related to the DID literature. Our main result generalizes Theorem 1 in de Chaisemartin and D’Haultfoeuille (2018). When the data has two groups and two periods, the Wald-DID estimand considered therein is equal to  $\beta_{fe}$  and  $\beta_{fd}$ . Our results on  $\beta_{fe}$  and  $\beta_{fd}$  are thus extensions of that theorem to the case with multiple periods and groups.<sup>3</sup> Moreover, our DID<sub>M</sub> estimator is related to the Wald-TC estimator with many groups and periods proposed in de Chaisemartin and D’Haultfoeuille (2018), and to the multi-period DID estimator proposed by Imai and Kim (2018). In Section 4, we explain the differences between those three estimators.

More recently, Borusyak and Jaravel (2017), Abraham and Sun (2018), Athey and Imbens (2018), Callaway and Sant’Anna (2018), and Goodman-Bacon (2018) study the special case of staggered adoption designs, where the treatment of a group is weakly increasing over time. Those papers derive some important results specific to that design that we do not consider here. Still, some of the results in those papers are related to ours, and we describe precisely those connections later in the paper. The most important dimension on which our paper differs from those is that our results apply to any two-way fixed effects regressions, not only to those with staggered adoption. In our survey of the AER papers estimating two-way fixed effects regressions, less than 10% have a staggered adoption design. This suggests that while staggered adoptions are an important research design, they may account for a relatively small minority of the applications where two-way fixed effects regressions have been used.

The paper is organized as follows. Section 2 introduces the set-up. Section 3 presents our decomposition results. Section 4 introduces our alternative estimator. Section 5 briefly describes some of the extensions covered in our Web Appendix. Section 6 presents our survey of the articles published in the AER, and our two empirical applications.

## 2 Set up

One considers observations that can be divided into  $G$  groups and  $T$  periods. For every  $(g, t) \in \{1, \dots, G\} \times \{1, \dots, T\}$ , let  $N_{g,t}$  denote the number of observations in group  $g$  at period  $t$ , and let  $N = \sum_{g,t} N_{g,t}$  be the total number of observations. The data may be an individual-level panel or repeated cross-section data set where groups are, say, individuals’ county of birth. The data could also be a cross-section where cohort of birth plays the role of time. For instance, Duflo (2001) compares the schooling of different cohorts in Indonesia, some of which were exposed to a school construction program. It is also possible that for all  $(g, t)$ ,  $N_{g,t} = 1$ , e.g. a group is one individual or firm. All of the above are special cases of the data structure we consider.

One is interested in measuring the effect of a treatment on some outcome. Throughout the paper

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<sup>3</sup>In fact, a preliminary version of our main result appeared in a working paper version of de Chaisemartin and D’Haultfoeuille (2018) (see Theorems S1 and S2 in de Chaisemartin and D’Haultfoeuille, 2015).

Here is some intuition as to why one weight is negative in this example. It follows from Equation (7) in the proof of Theorem 1 (see also Theorem 1 in Goodman-Bacon, 2018) that in this simple example,  $\beta_{fe} = (\text{DID}_1 + \text{DID}_2)/2$ , with

$$\begin{aligned}\text{DID}_1 &= E(Y_{2,2}) - E(Y_{2,1}) - (E(Y_{1,2}) - E(Y_{1,1})), \\ \text{DID}_2 &= E(Y_{1,3}) - E(Y_{1,2}) - (E(Y_{2,3}) - E(Y_{2,2})).\end{aligned}$$

The first DID compares the evolution of the mean outcome from period 1 to 2 in group 2 and in group 1. The second one compares the evolution of the mean outcome from period 2 to 3 in group 1 and in group 2. The control group in the second DID, group 2, is treated both in the pre and in the post period. Therefore, under the common trends assumption, it follows from Lemma 1 in Appendix A (a similar result appears in Lemma 1 of de Chaisemartin (2011) and in Equation (13) of Goodman-Bacon (2018)) that  $\text{DID}_1 = E[\Delta_{2,2}]$ , but

$$\text{DID}_2 = E[\Delta_{1,3}] - (E[\Delta_{2,3}] - E[\Delta_{2,2}]).$$

$\text{DID}_2$  is equal to the ATE in group 1 period 3, minus the change in group 2's ATE between periods 2 and 3. Intuitively, the mean outcome of groups 1 and 2 may follow different trends from period 2 to 3 either because group 1 becomes treated, or because group 2's ATE changes. The intuition that negative weights arise because  $\hat{\beta}_{fe}$  uses treated observations as controls also appears in Borusyak and Jaravel (2017).

We now generalize the previous illustration by characterizing the  $(g, t)$  cells whose ATEs are weighted negatively by  $\beta_{fe}$ .

**Proposition 1** *Suppose that Assumption 1 holds and for all  $t \geq 2$   $N_{g,t}/N_{g,t-1}$  does not vary across  $g$ . Then, for all  $(g, t, t')$  such that  $D_{g,t} = D_{g,t'} = 1$ ,  $D_{.,t} > D_{.,t'}$  implies  $w_{g,t} < w_{g,t'}$ . Similarly, for all  $(g, g', t)$  such that  $D_{g,t} = D_{g',t} = 1$ ,  $D_{g,.} > D_{g',.}$  implies  $w_{g,t} < w_{g',t}$ .*

Proposition 1 shows that  $\beta_{fe}$  is more likely to assign a negative weight to periods where a large fraction of groups are treated, and to groups treated for many periods. Then, negative weights are a concern when treatment effects differ between periods with many versus few treated groups, or between groups treated for many versus few periods.

Proposition 1 has interesting implications in staggered adoption designs, a special case of sharp designs defined as follows.

**Assumption 6** (*Staggered adoption designs*) *For all  $g$ ,  $D_{g,t} \geq D_{g,t-1}$  for all  $t \geq 2$ .*

Assumption 6 is satisfied in applications where groups adopt a treatment at heterogeneous dates (see e.g. Athey and Stern, 2002). In that design, Borusyak and Jaravel (2017) show that  $\beta_{fe}$  is more likely to assign a negative weight to treatment effects at the last periods of the panel.